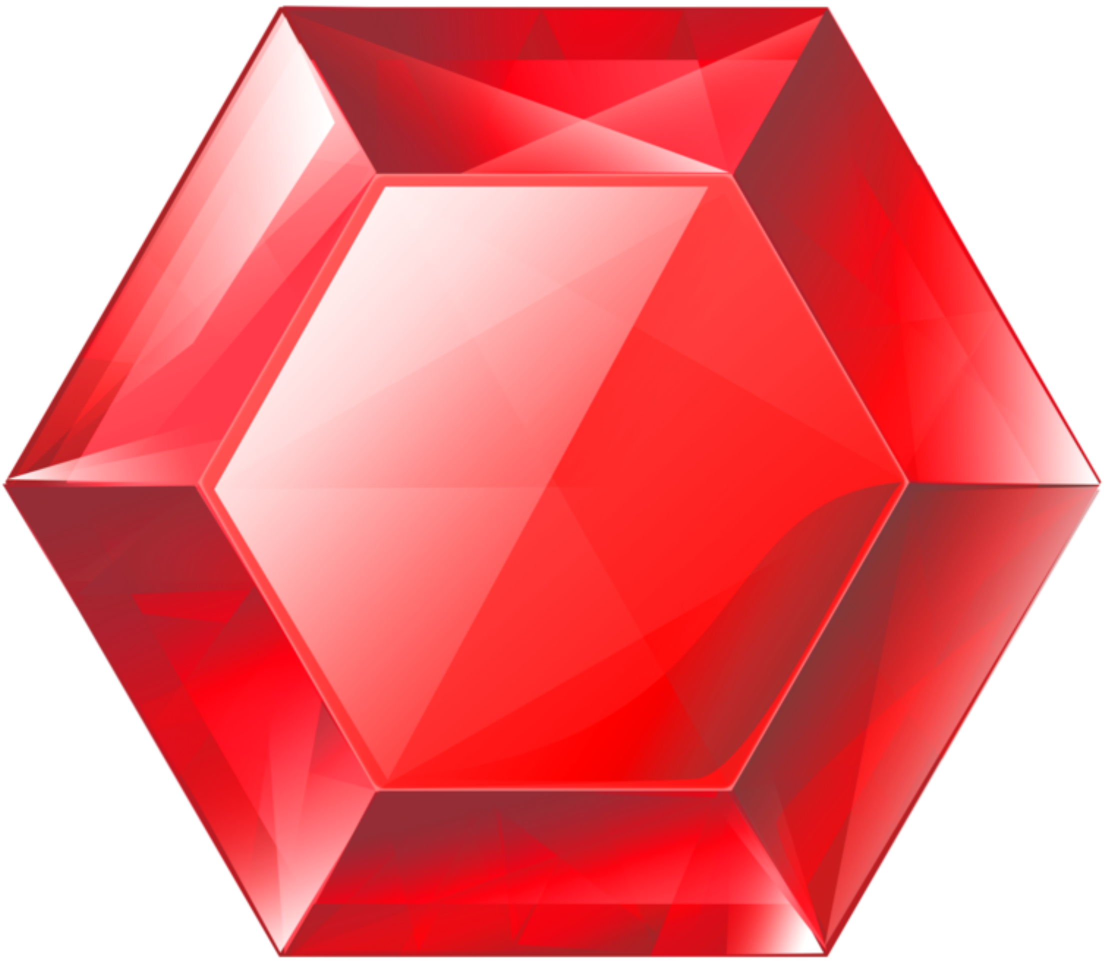
gemstone-rs VS Code Workbench

Sidebar browsing, codegen, and explorer launch workflows



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VS Code Workbench

`gemstone-rs Workbench` is the VS Code companion for the Rust CLI and local explorer. It keeps the CLI as the stable contract and adds a sidebar, output panel, generated-file previews, and codegen diff prompts.

Marketplace:

```
https://marketplace.visualstudio.com/items?itemName=unicompute.gemstone-rs-workbench
```

Setup

For installed CLI tools:

```
{
  "gemstoneRs.cliPath": "gemstone-rs",
  "gemstoneRs.explorerPath": "gemstone-rs-explorer",
  "gemstoneRs.codegenConfig": "gemstone-rs.codegen"
}
```

For a source checkout:

```
{
  "gemstoneRs.checkoutPath": "/path/to/gemstone-rs",
  "gemstoneRs.useCargo": true,
  "gemstoneRs.codegenConfig": "examples/codegen/gemstone-rs.codegen"
}
```

Use the same `GS_*` environment as the CLI.

Sidebar Tree

Open the GemStone RS activity bar item.

The sidebar contains:

- `Dictionaries`
- `Codegen Config`
- `Explorer`

`Dictionaries` expands live dictionaries, classes, protocols, and methods. Selecting a method opens its source in an untitled Smalltalk editor.

`Codegen Config` exposes:

- Discover from Live Stone
- Generate Mapping Config
- Preview BridgeRoot
- List BridgeRoot Keys
- Put BridgeRoot String
- Remove BridgeRoot Key
- Run Generated Mapping Example
- Preview Wrappers
- Diff Generated Output
- Check Freshness
- Generate Wrappers
- Open Codegen Docs

`Explorer` exposes:

- Doctor
- Verify Setup
- Eval Smalltalk
- Launch Explorer

Command Palette

Commands:

- `GemStone RS: Verify Setup`
- `GemStone RS: Doctor`
- `GemStone RS: Eval Smalltalk`
- `GemStone RS: Browse Dictionaries`
- `GemStone RS: Browse Classes`
- `GemStone RS: Codegen Init`
- `GemStone RS: Codegen Discover`
- `GemStone RS: Generate Mapping Config`
- `GemStone RS: Preview BridgeRoot`
- `GemStone RS: List BridgeRoot Keys`
- `GemStone RS: Put BridgeRoot String`
- `GemStone RS: Remove BridgeRoot Key`
- `GemStone RS: Run Generated Mapping Example`
- `GemStone RS: Codegen Preview`
- `GemStone RS: Codegen Diff`
- `GemStone RS: Codegen Check`
- `GemStone RS: Codegen Generate`

- `GemStone RS: Launch Explorer`
- `GemStone RS: Open Method Source`
- `GemStone RS: Open Codegen Docs`

`GemStone RS: Verify Setup` and `GemStone RS: Doctor` both run the CLI `gemstone-rs doctor` report, so the workbench setup view and terminal setup use the same diagnostic path.

Codegen Workflow

1. Set `gemstoneRs.codegenConfig`.
2. Run `GemStone RS: Codegen Discover` or edit the config manually.
3. Run `GemStone RS: Codegen Preview`.
4. Run `GemStone RS: Codegen Diff`.
5. Run `GemStone RS: Codegen Generate`.

`Codegen Generate` runs the diff first. If output would change, it opens the diff and asks before writing.

Object Mapping Workflow

Use `GemStone RS: Generate Mapping Config` when you want the live stone to inspect a GemStone class and propose a `BridgeMapped` config. The command asks for:

- config path
- Rust struct name
- GemStone class reference, such as `Object` or `UserGlobals:OkzBooking`

Use `GemStone RS: Preview BridgeRoot` after starting the local explorer. It opens:

```
http://127.0.0.1:8787/api/bridge/root
```

Use `GemStone RS: List BridgeRoot Keys` for the same inspection path without opening a browser; it runs `gemstone-rs bridge keys` and writes the key OOPs, class OOPs, `printString` values, and identity ids to the output panel.

Use `GemStone RS: Put BridgeRoot String` when you want a quick live-write smoke test from VS Code. It asks for a key and string value, then runs:

```
gemstone-rs bridge put <key> <value> --type String
```

Use `GemStone RS: Remove BridgeRoot Key` to remove one BridgeRoot key after a confirmation prompt:

```
gemstone-rs bridge remove <key>
```

Use `GemStone RS: Run Generated Mapping Example` to run:

```
cargo run -p gemstone-rs --example generated_mapping_app
```

Develop the Extension

```
cd vscode-gemstone-rs-workbench
npm ci
npm run check
npm run package -- --out gemstone-rs-workbench-0.2.4.vsix
```

From the repository root:

```
make vscode-package
```

Later Feature Work

Embedding `gemstone-rs-explorer` as a VS Code webview should remain a feature release because it changes the user experience and test surface. The current extension intentionally launches the explorer externally and opens the local URL.

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